



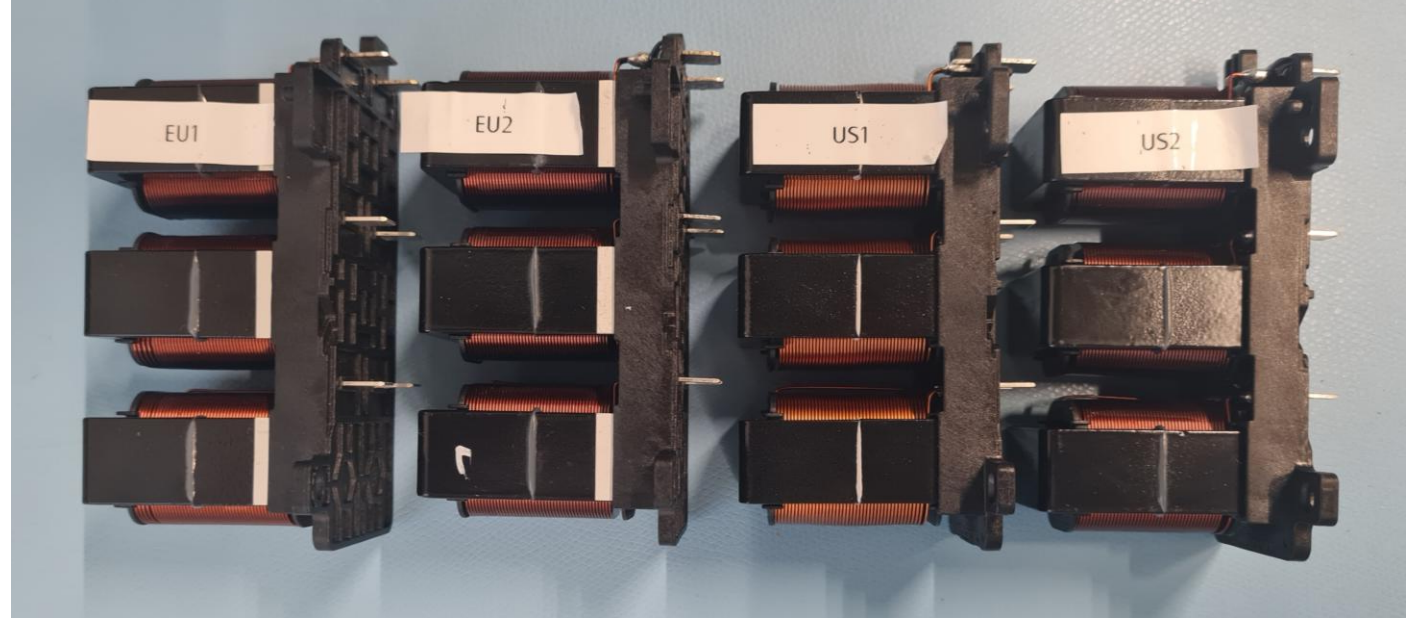
Alternative measurement with Precision LCR meter instead of VNA

Kardos Milán, EA-EMC, 2025.07.02

DUTs

General informations

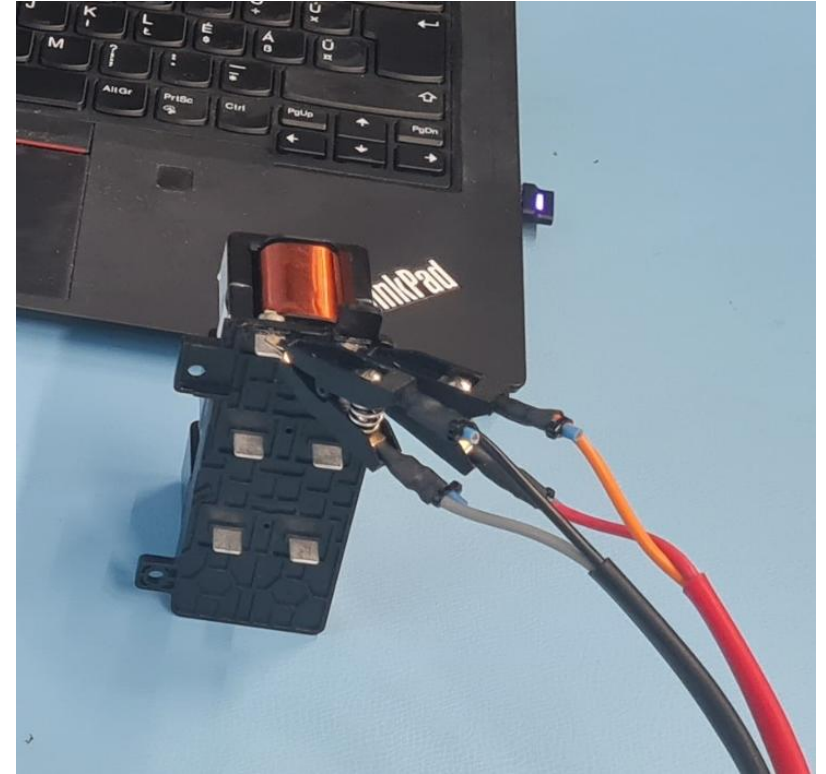
- Four coils was measured
 - EU1
 - EU2
 - US1
 - US2
- In each measurement the top coil was measured with VNA and also with LCR



LCR meter

Settings and cautions

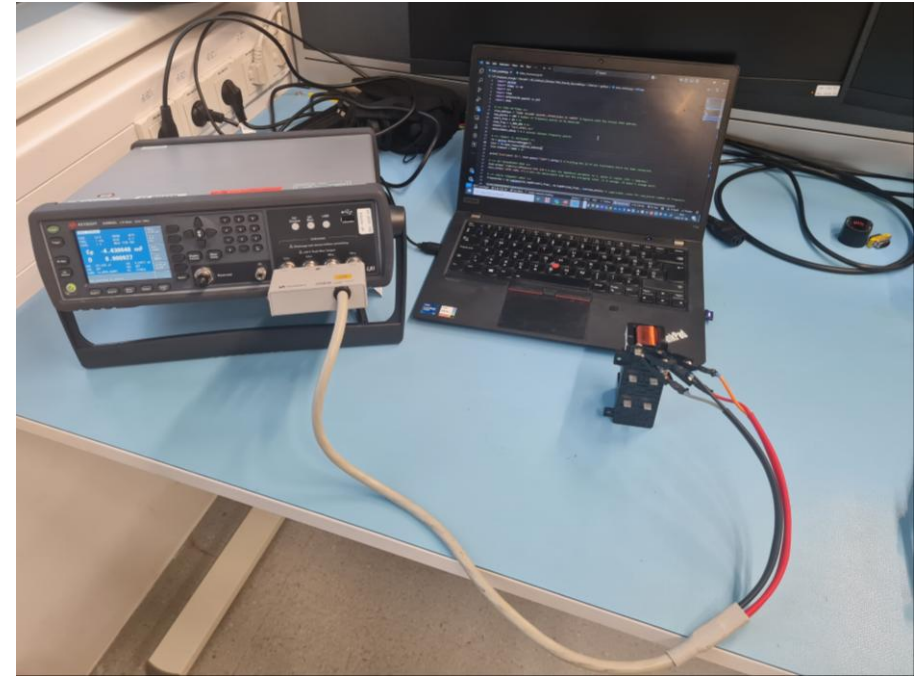
- LCR meter range: 20Hz – 1MHz
- Test signal: 1V
- Tested corrections: 20Hz, 500kHz, 1MHz
- If you are using **clip-type** equipment as the pictures show take care of the right contact it can easily cause **unreliable measurements!**



LCR meter

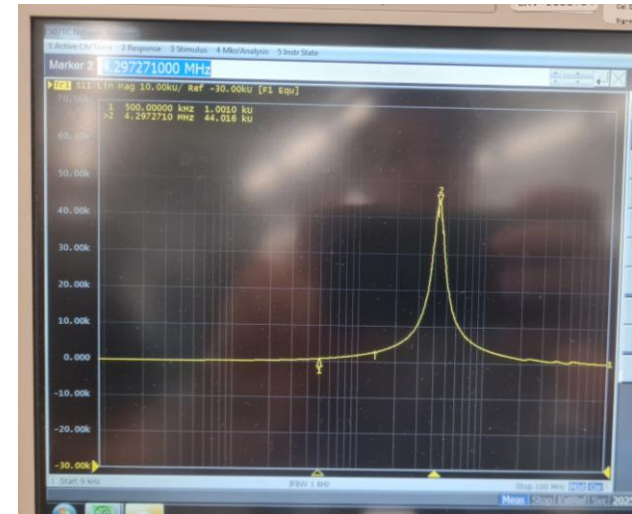
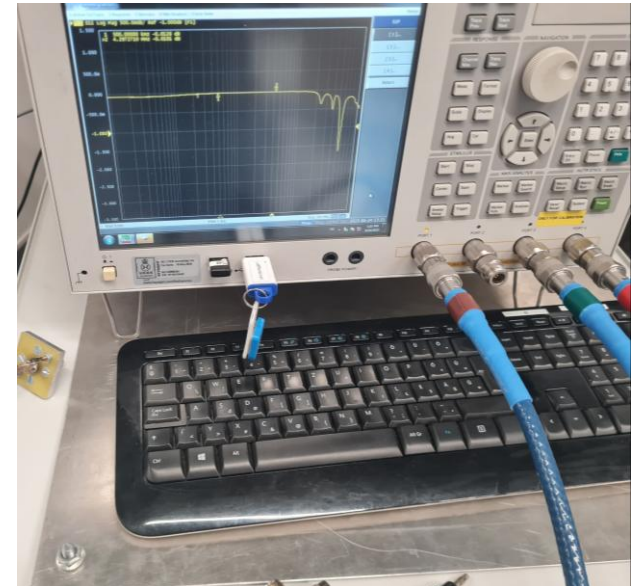
Measurement automation (Python)

- The automation requirements are:
 - PC USB right
 - Python kernel
 - Program to run python code (in this case Visual Studio Code)
- Setting up the automated measurement:
 - Define instrument visa address (read it with NI max or other instrument control software or manually from the device)
 - Give the number of frequency points
 - Define start and stop frequency
 - Define output CSV name
 - Define the measurement delay (1s is recommended)
 - Plug in the USB cable and run the code
- You may have to install additional packages if python requires it!
 - Do it within the code with the following command: `pip install packagename`



VNA Settings

- VNA range: 9kHz – 100MHz
- Frequency for verification: 500kHz
- You should save the S1P parameters to USB drive and process the data (calculate impedances) later with the python script



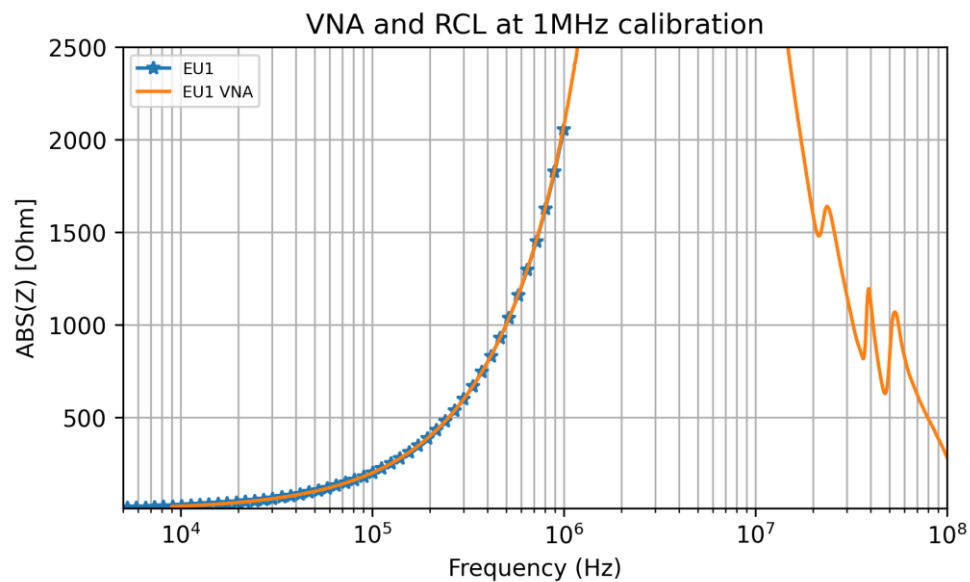
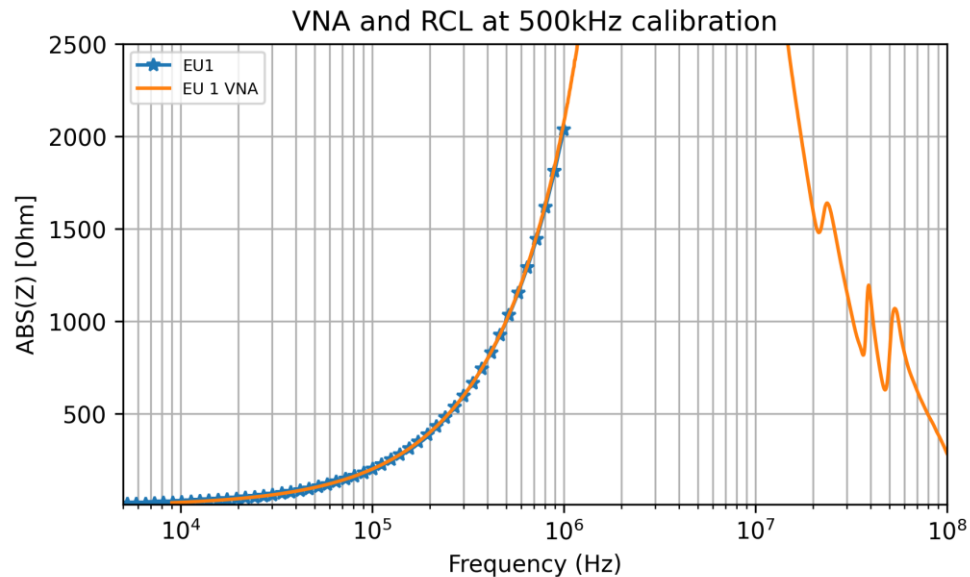
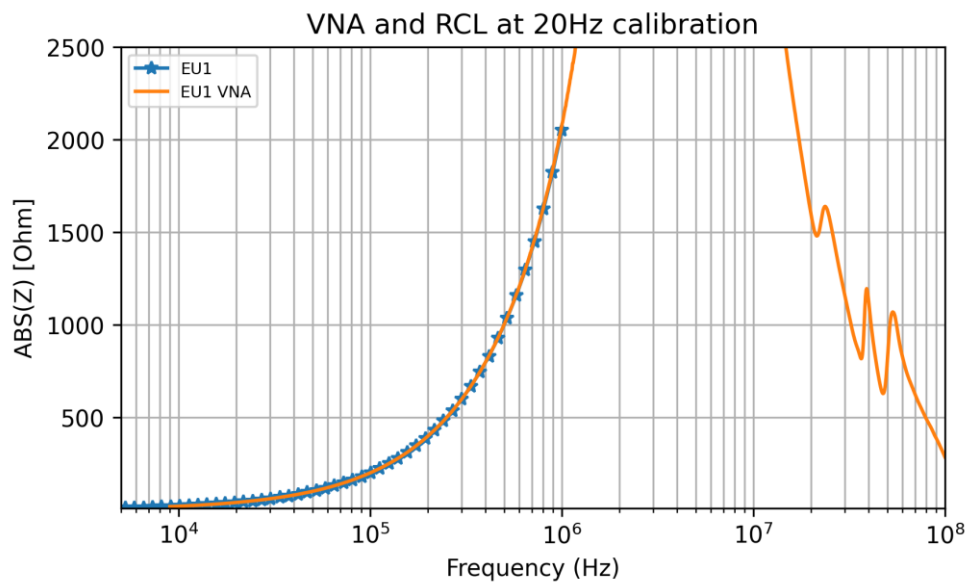
Evaluation

Python code

- The evaluation and the plots were made with the python plot: [Milan_business.ipynb](#)
- The evaluation was according to CISPR17
- The comments in the source code explains the usage
- The code can be used to evaluate S2P and S1P parameters and CSV-s made by the RCL instrument control python code too
- Some customization in the code may be needed! If so please make a new code cell and run it separately!

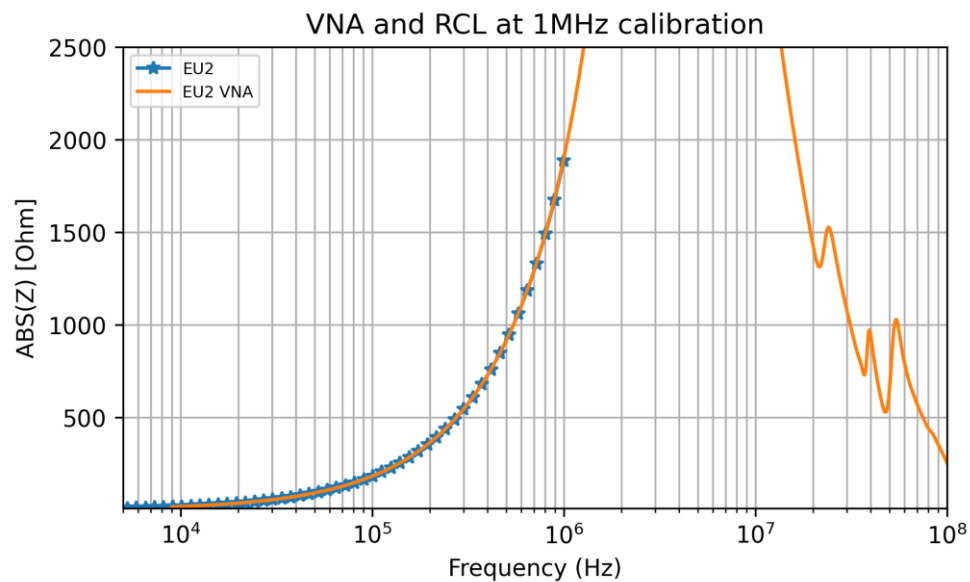
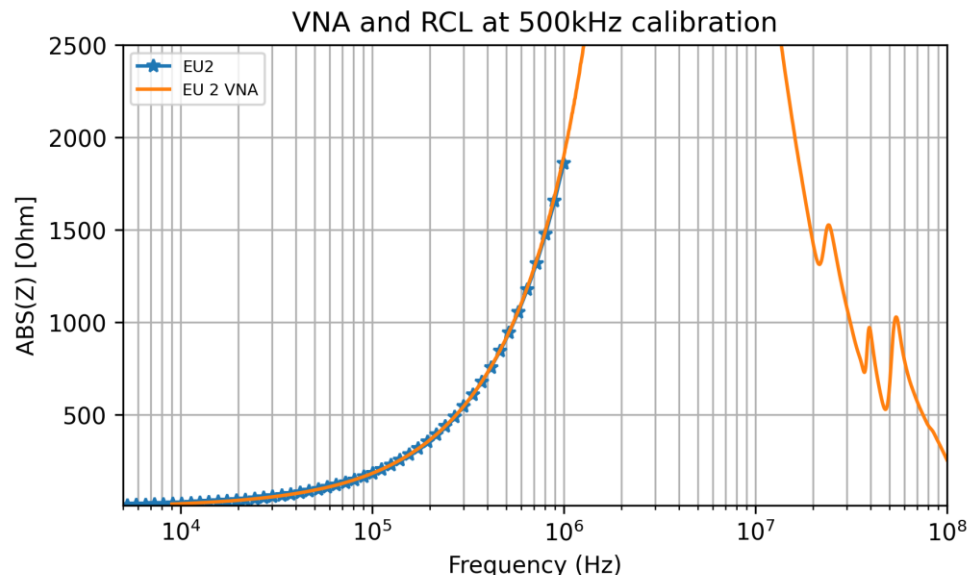
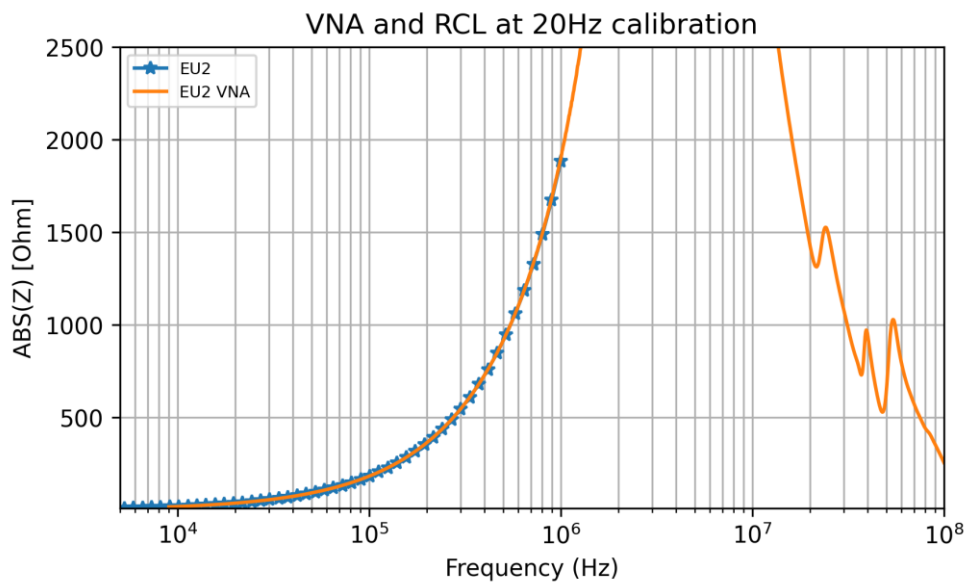
RCL meter measurements

EU1



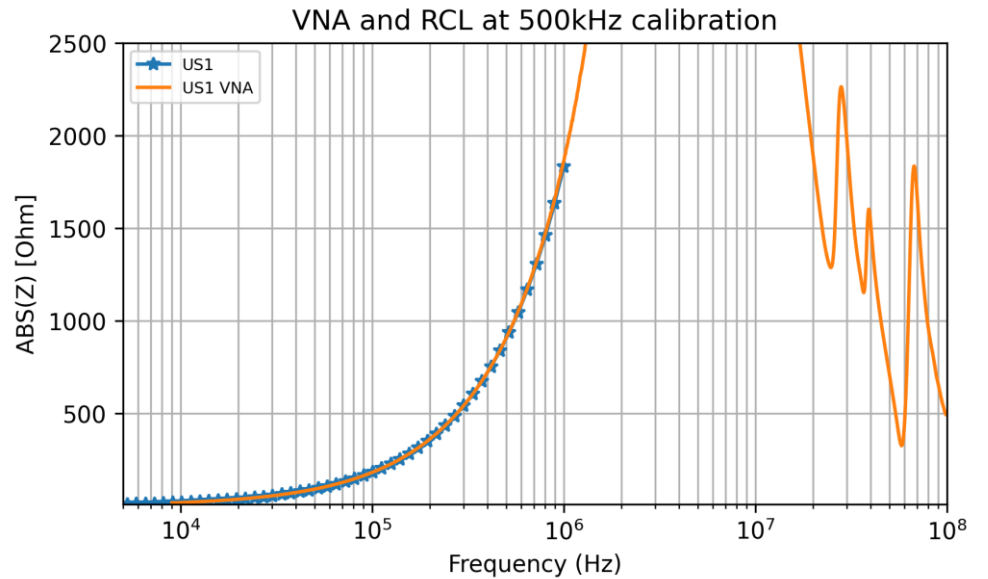
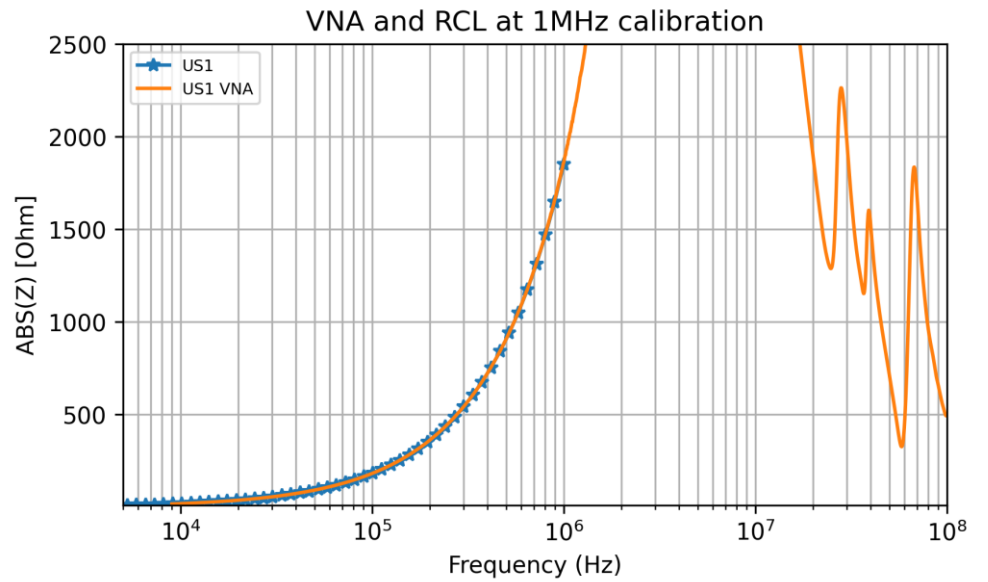
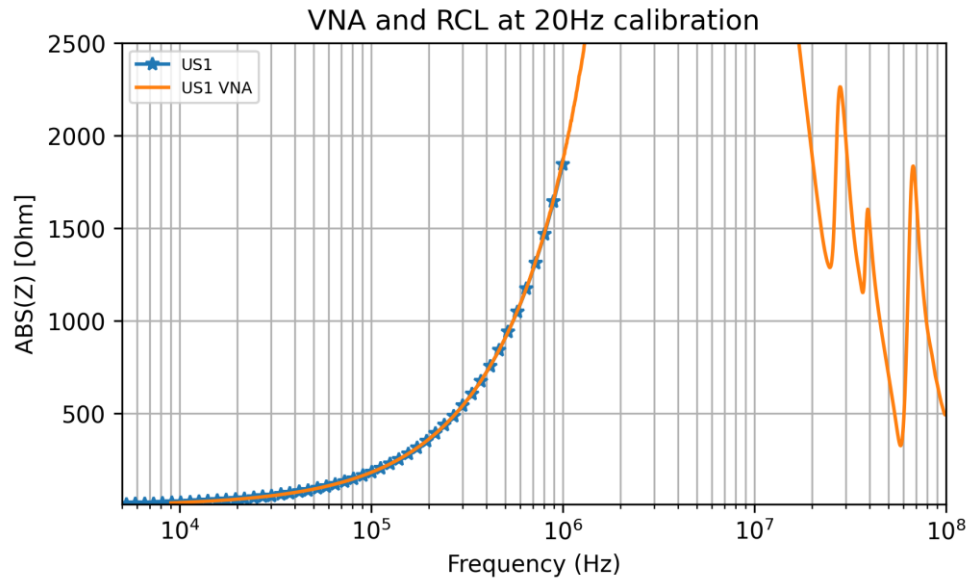
RCL meter measurements

EU2



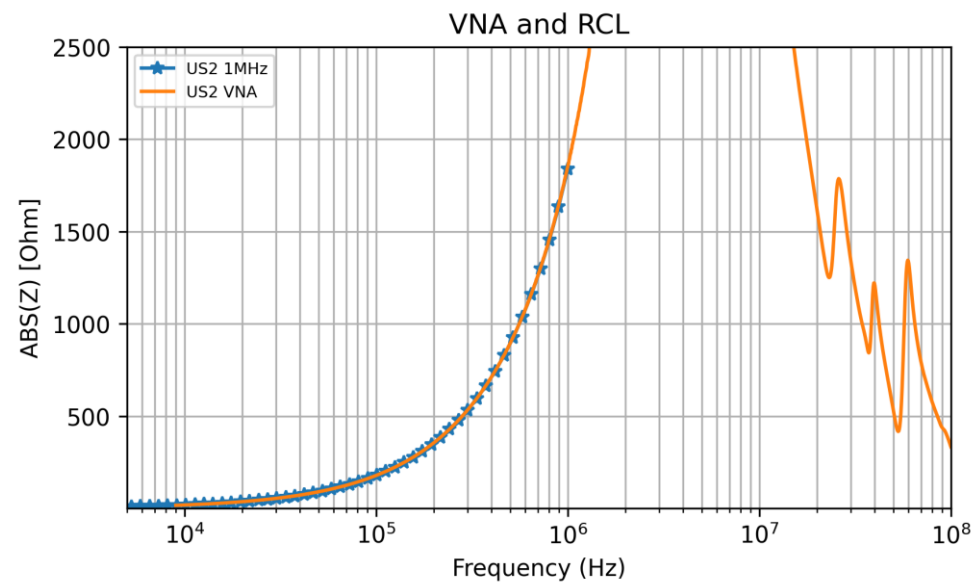
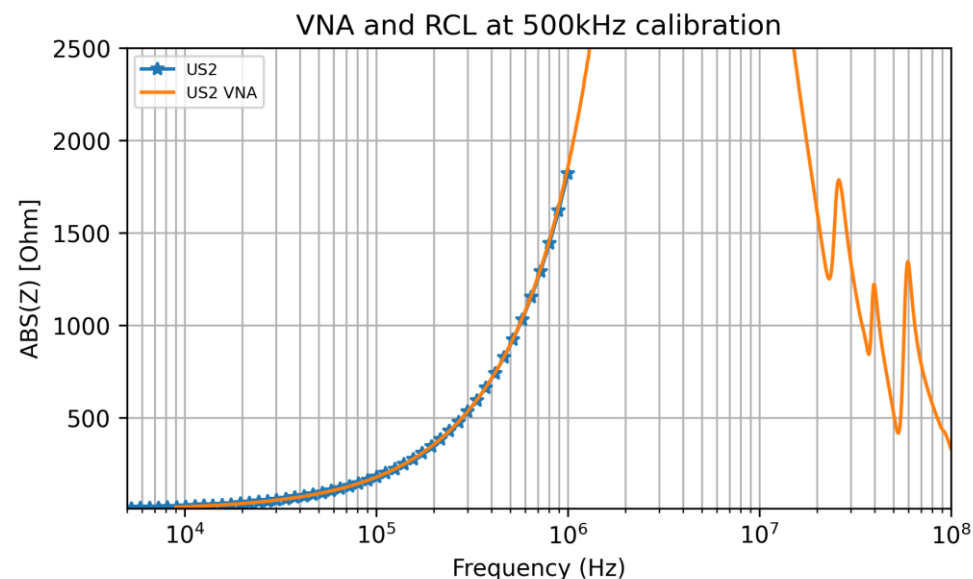
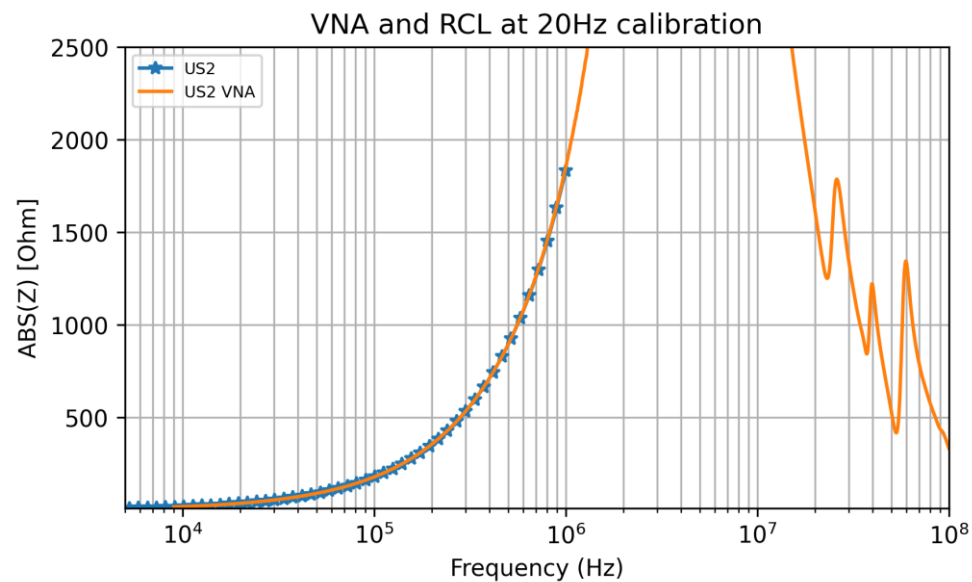
RCL meter measurements

US1



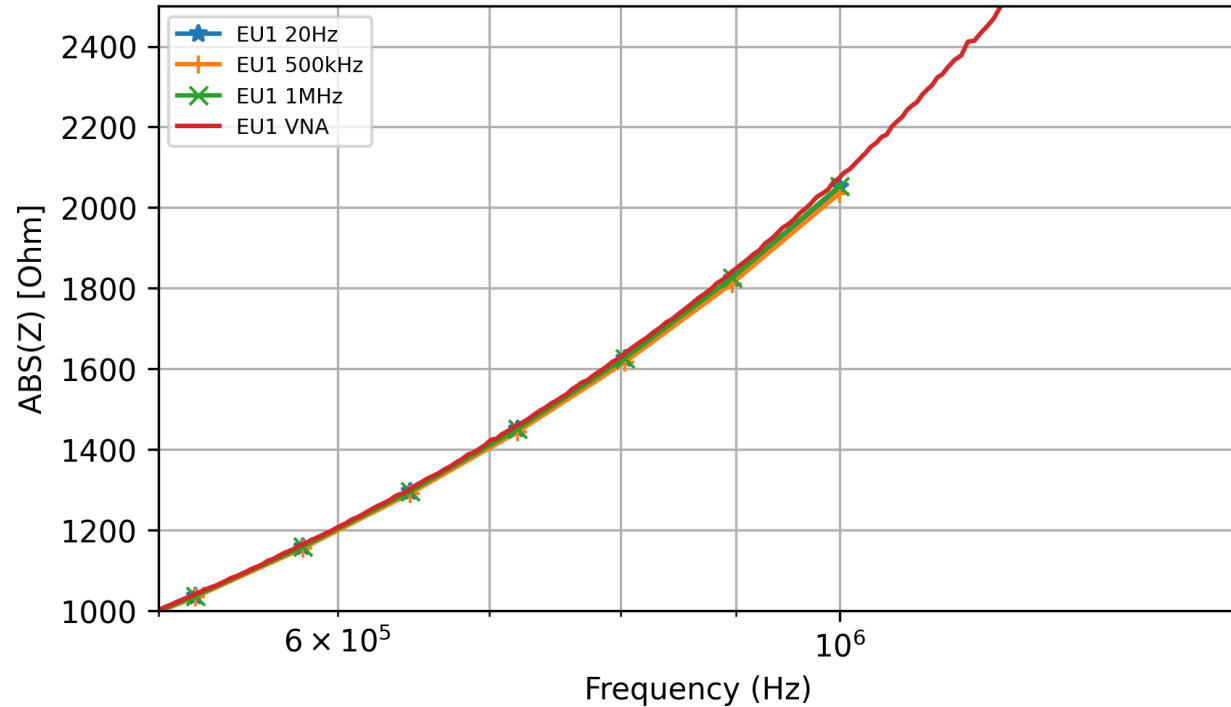
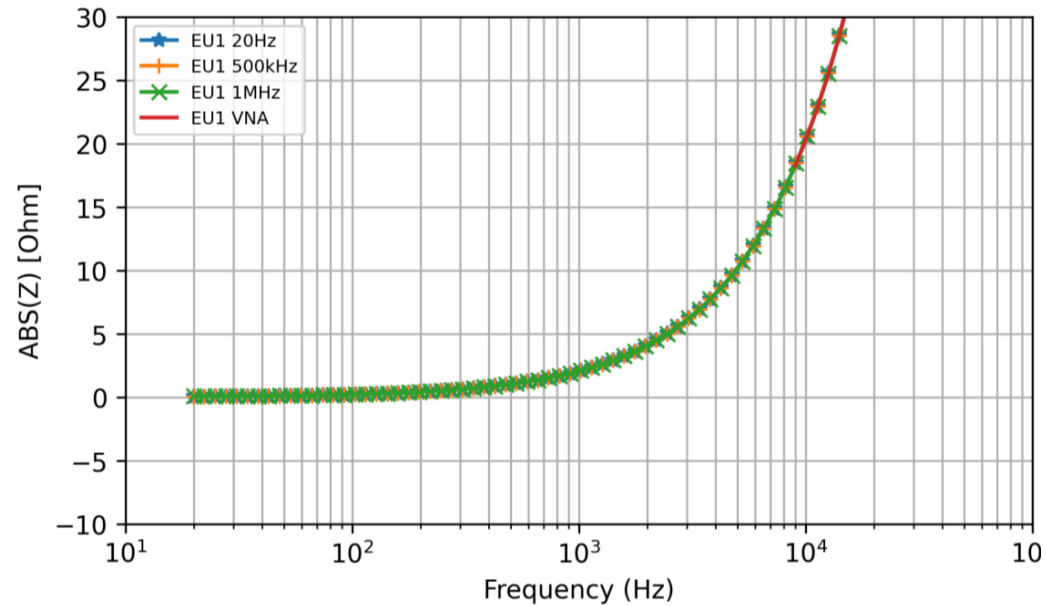
RCL meter measurements

US2



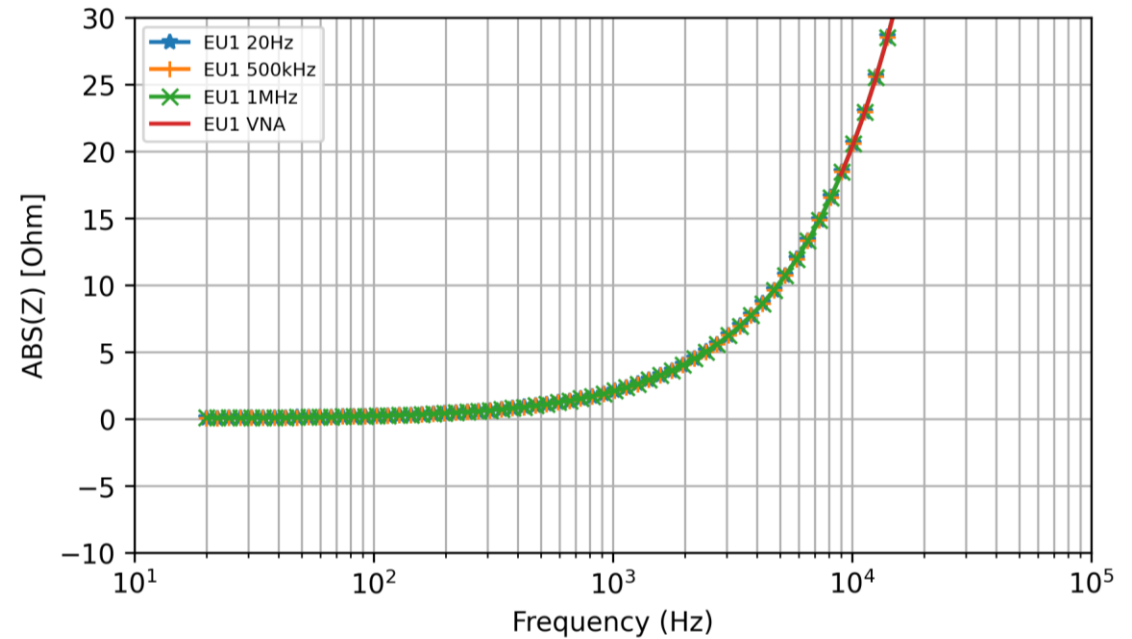
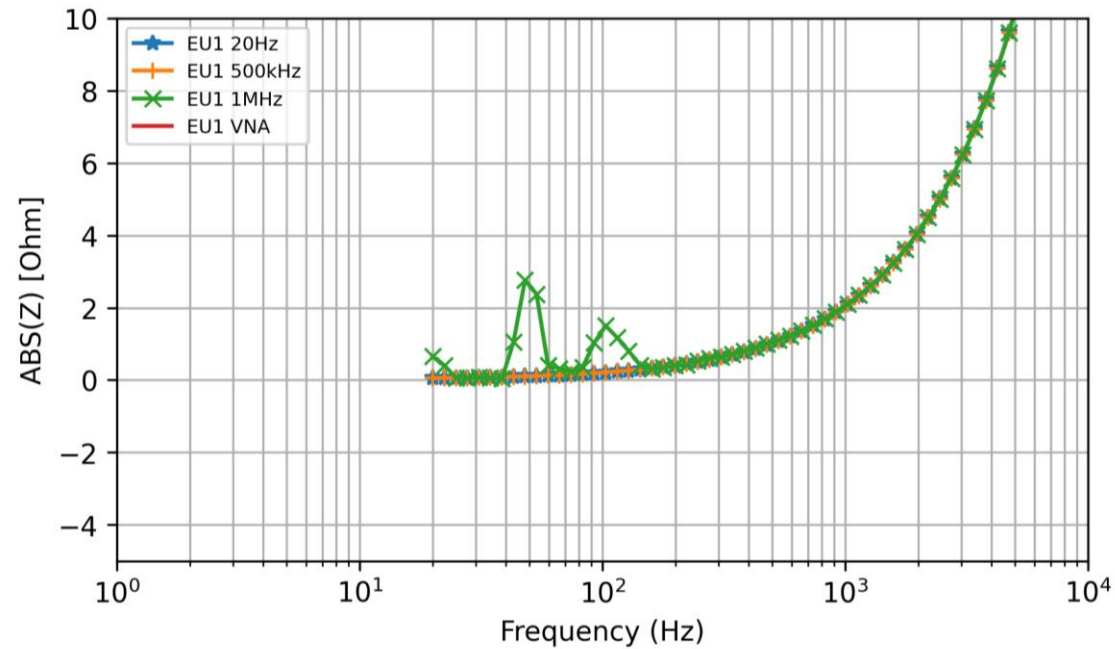
RCL meter measurements

EU1 compare corrections



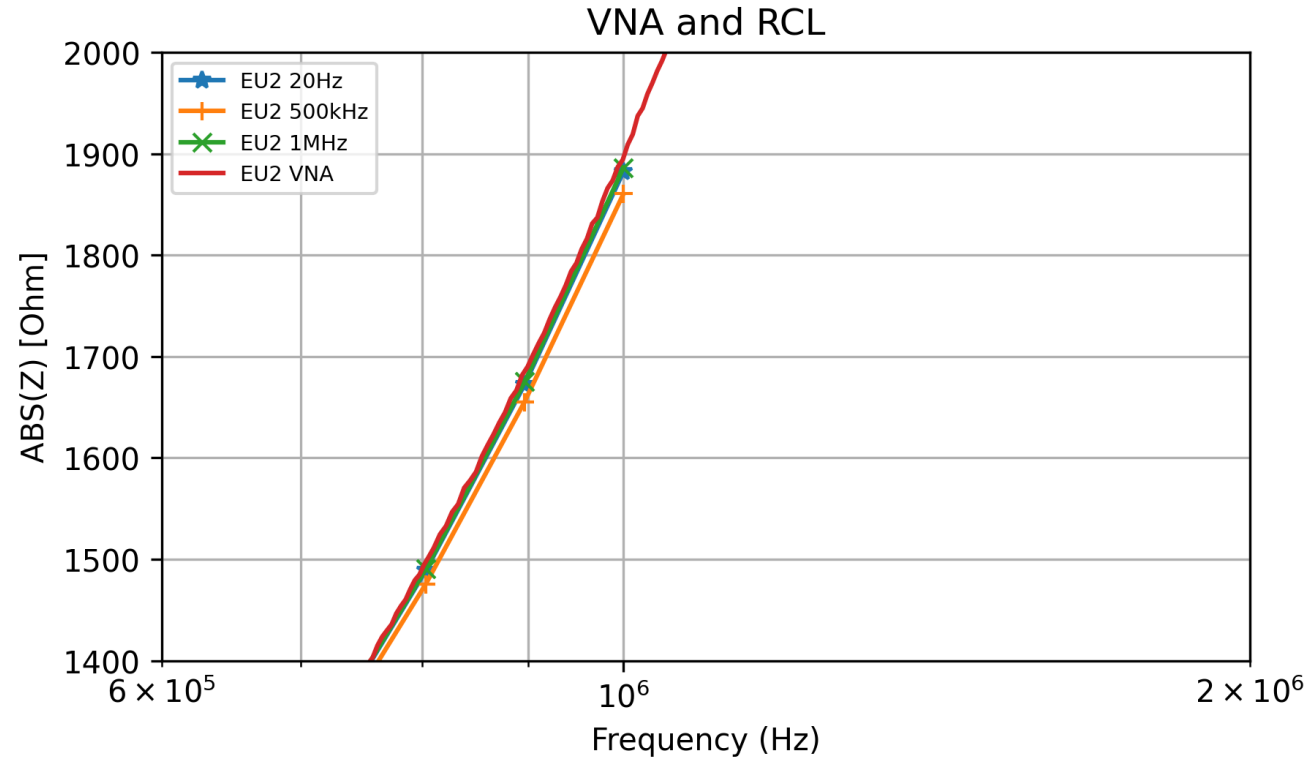
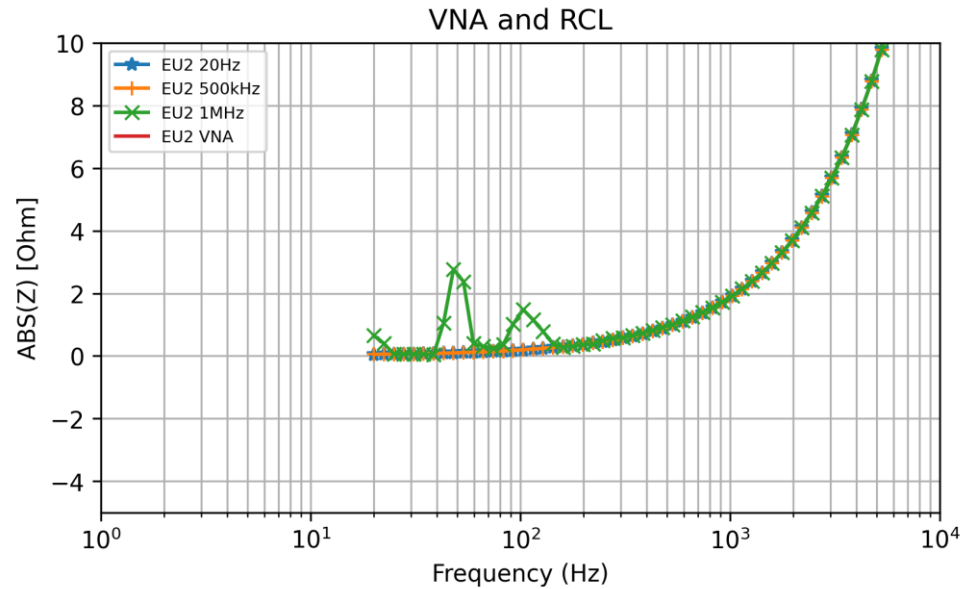
RCL meter measurements

EU1 failed measurement



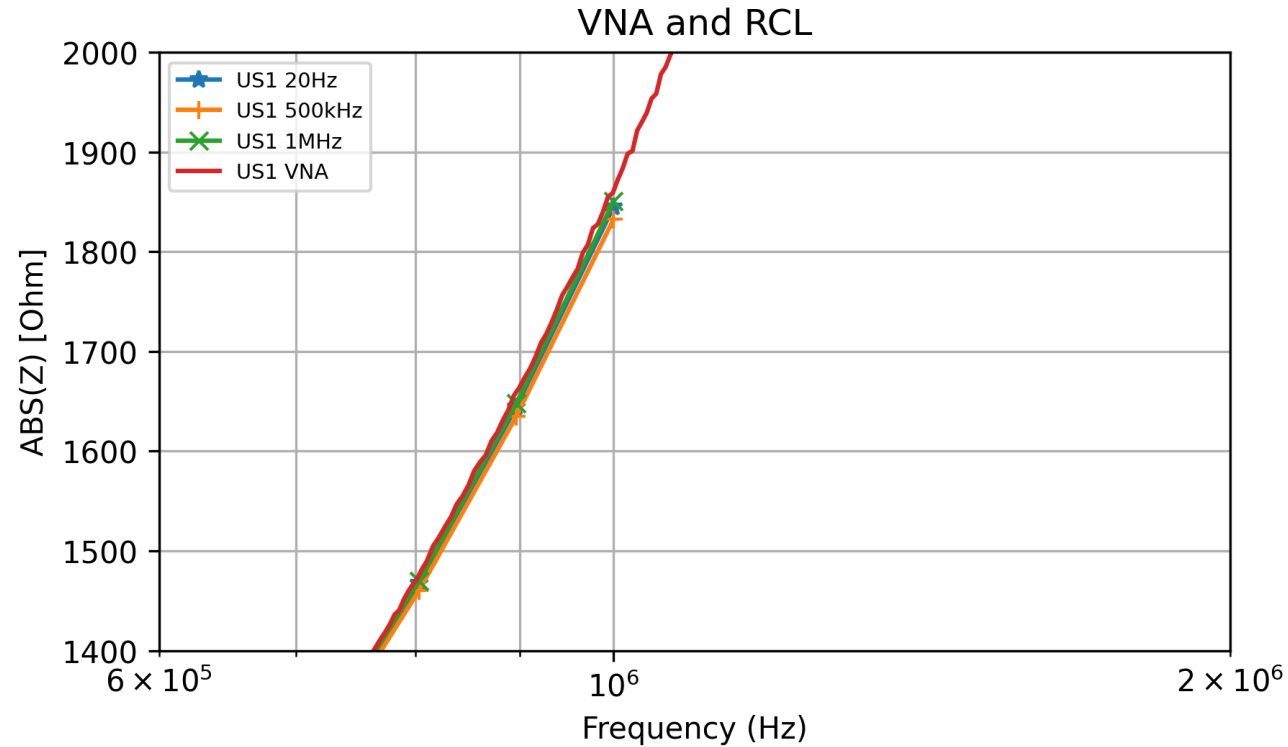
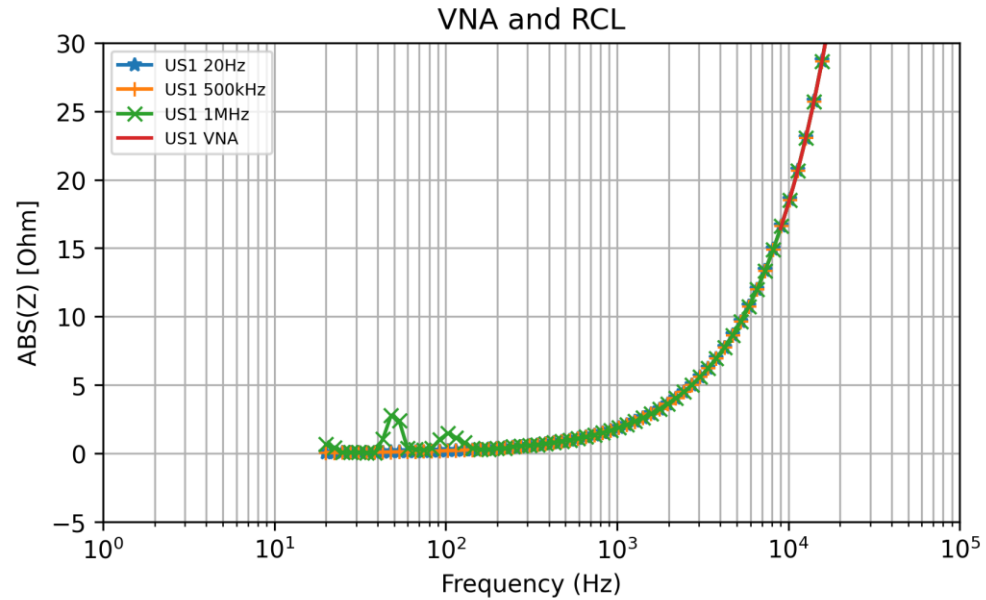
RCL meter measurements

EU2 compare corrections



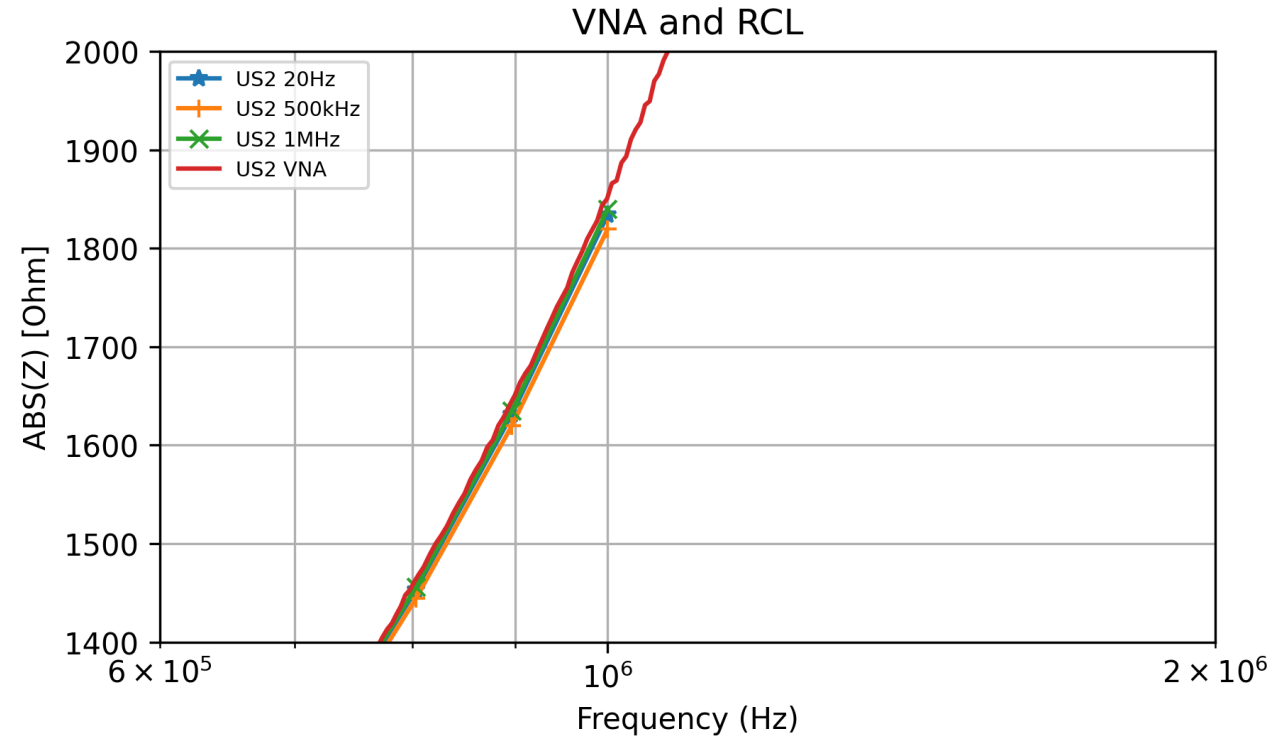
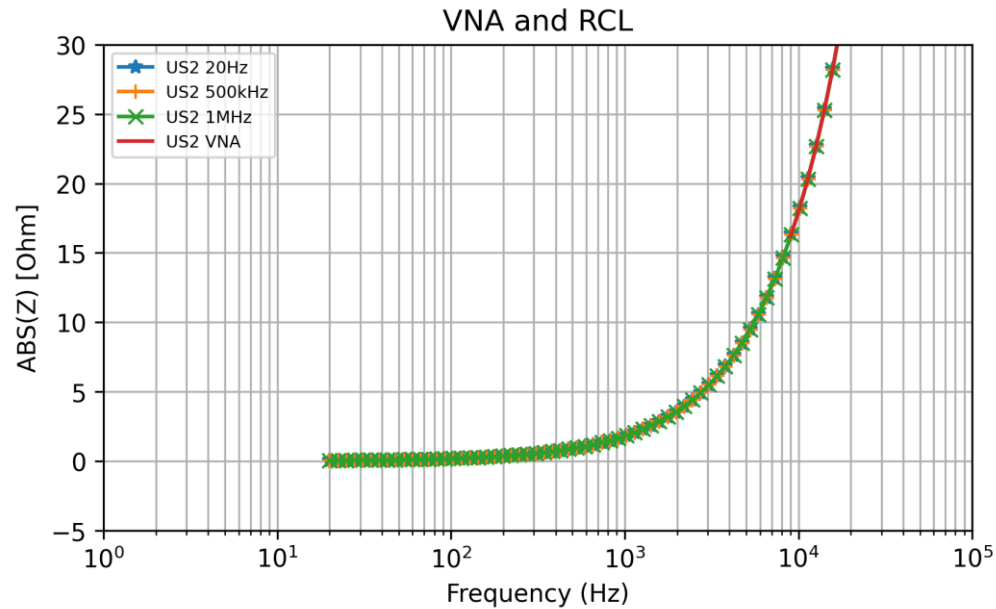
RCL meter measurements

US1 compare corrections



RCL meter measurements

US2 compare corrections



RCL meter measurements

Conclusions

- The worst case difference between the VNA and RCL measurements is around 20 Ohm at 1 MHz, which is **less than 2%** measurement uncertainty (in fact it's around 1.2%)
- RCL meter can be used at lower frequencies instead of VNA
- The biggest deviations in impedance is at around 1 MHz
- We have to take care of the right connection of the RCL clamps to have correct measurements
- The majority of false measurements are at low frequencies

RCL meter measurements

Conclusions

- With Bence's python script the method is tested and it can be directly launched from your PC via USB
- There are very slight differences between the corrections, the 1MHz seems to have the best behaviour the 20Hz follows it and at the end the „worst” is the 500kHz.
- Links to python codes:
 - RCL measurement automatization:
S:\07_Employee_storage\Bernath\40_CoilAnal_LCRmeter-VNA_Swordy_GyurosBenya\Charcon\python\LCR_control.py
 - RCL and S1P evaluation:
S:\07_Employee_storage\Bernath\33_sci_kit_RF_learning\01_Milan_fancy_stuff\Milan_business.ipynb

